OpenMagnetics

Steps

Design Req. ↓ Op. Points ↓ Magnetic Builder ↓ Summary

Tool menu

Magnetic Summary

MAS Exports

Core Exports

Coil Exports

Circuit Simulators Exports

Core data

Core Shape

Name	ETD 59/31/22
No. Stacks	1
Effective length	0.14 m
Effective area	367.98 mm^2
Effective volume	52641.39 mm^3
Minimum Area	366.21 mm^2

Core Gapping

Type	Length	Height offset
Residual	5 µm	0 m
Residual	5 µm	0 m
Residual	5 µm	0 m

Material Parameters

Name	N87	
Manufacturer	TDK	
	25°C	100°C
Permeance (AL value)	6.42 $\mu H/(tu.)^2$	10.07 $\mu H/(tu.)^2$
Initial Permeability (μ_i)	2304	3983
Eff. Permeability (μ_e)	1984	3117
Resistivity	10 Ωm	10 Ωm
Curie Temperature	210 °C	
Density	4.85 Mg/m^3	

Coil Global Parameters

No. Sections	5
No. Layers	15

Primary

No. turns	3
No. parallels	1
Wire	Litz Round 3.15 - Grade 1

Secondary

No. turns	26
No. parallels	1
Wire	Litz undefined

Simulation result per Operating Point

Op. Point No. 1

Core

Mag. Ind.	90.7 µH
Core losses	3.37 W
Core temp.	120.99 °C
B peak	0.11 T

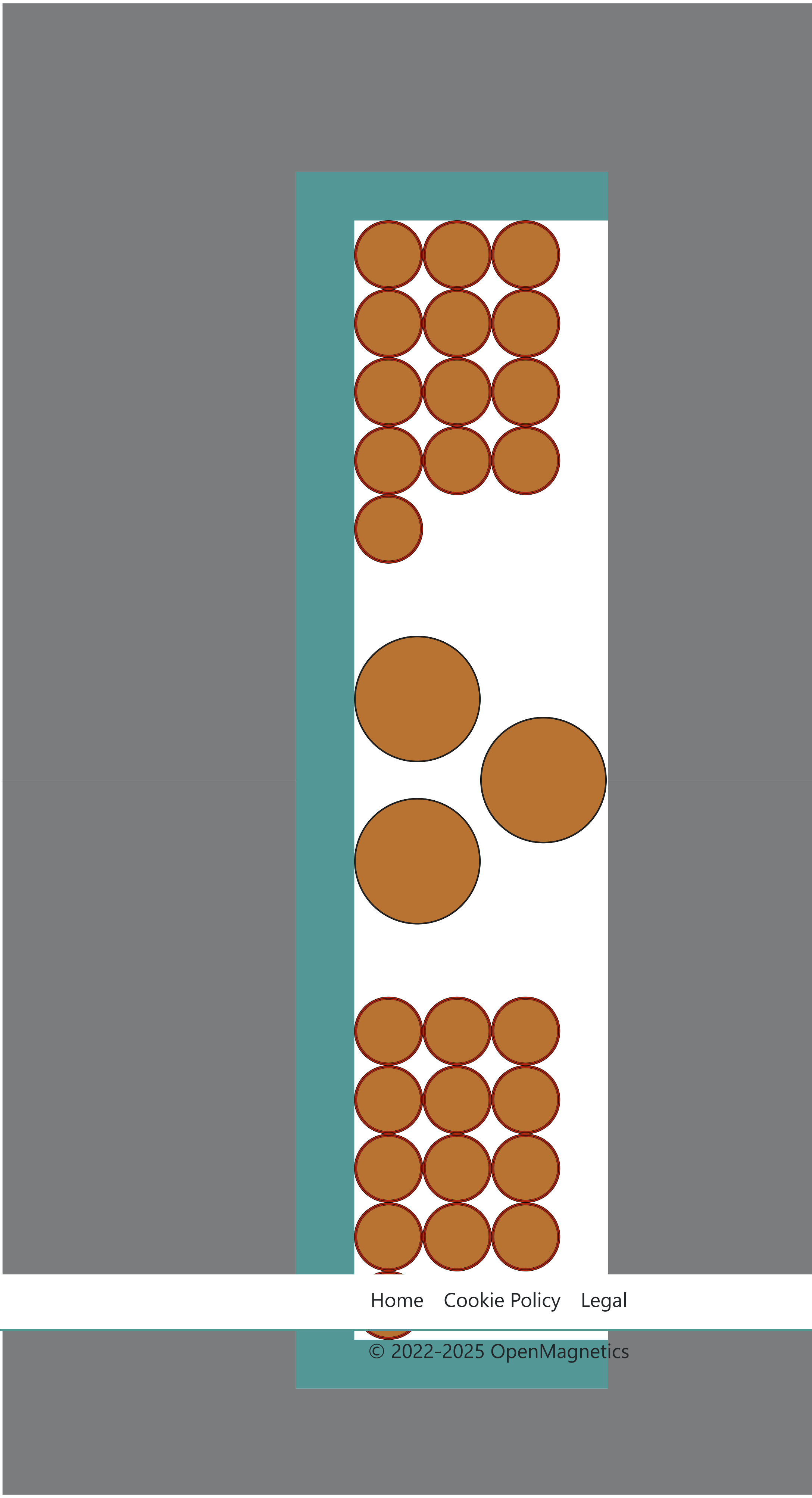
Coil

Winding	DC Res.	Curr. Density	Wind. Loss	Leak. Ind.
Primary	898.72 µΩ	15.18 A/mm²	14.05 W	
Secondary	30.64 mΩ	5.56 A/mm²	4.35 W	742.46 nH

Coil Losses Breakdown

Winding	Ohmic Loss	Skin Loss	Prox. Loss
Primary	12.77 W	1 mW	1.28 W
Secondary	3.65 W	9.59 µW	0.7 W

Core Coil



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